Statement of Work (SOW)

AI-Driven Static Loaded Radius (SLR) Prediction Using Historical Tyre Data

Industry Partner: CEAT Specialty Tyres Ltd.

Institute: Vishwakarma Institute of Technology, Pune

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Project Start Date: 12 August 2026

Target Completion Date: 31 October 2026

Project Duration: Approximately 12 weeks

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1. Project Objective

The objective is to develop a CAMSO mobile application for Android and iOS that provides centralized access to product information, technical resources, customer support, and distributor services through a single digital platform.

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2. Scope of Work

The project will cover the complete workflow from CAMSO mobile application development and product discovery to customer support, distributor services, advanced features, testing, and production deployment across Android and iOS.

The major scope areas are:

1. Product catalogue and product information management for Compact Construction Tires and Tracks.
2. Product search and filtering based on product name, PID, vehicle, equipment model, application type, and industry segment.
3. Customer support and engagement including Support Library, FAQs, testimonials, VOC feedback, bookmarks, warranty information, and product updates.
4. Multilingual support and user management including English, French, German, Italian, login, profiles, and language preferences.
5. Dealer and distributor services including dealer discovery, distributor login, role-based access, assets, training, and claims.
6. Technical tools and advanced features including Product Selector, Pressure Advisor, TKPH Selector, availability, voice/image search, certifications, and pricing updates.
7. Calculators and technical decision-support features including LOCS and MHE Intensity calculators, subject to evaluation.
8. Admin CMS and analytics for content management, translations, campaigns, product searches, downloads, bookmarks, and user interests.
9. Backend, API, Firebase, and application integration supporting the mobile application and its services.
10. Testing, production deployment, documentation, and knowledge transfer for the complete application across the three planned launches.

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3. Final Deliverables

1. Cross-platform CAMSO Mobile Application for Android and iOS covering Compact Construction Tires and Tracks.
2. Product Catalogue and Search Module for browsing products, specifications, filters, product IDs and equipment/application-based search.
3. Support and Content Module providing catalogues, brochures, manuals, technical documents, FAQs, testimonials and offline-accessible resources.
4. User Engagement Features including bookmarks, VOC feedback, product updates, push notifications and multilingual support.
5. User Login and Distributor Module providing profile management, role-based distributor access, marketing assets and distributor-specific content.
6. Training and Claims Module for distributor training content, claim registration, document/image uploads and claim-status tracking.
7. Product Selectors and Tools including Product Selector, Pressure Advisor, Reverse Track Selector, Rubber Pad Selector and other approved tools.
8. Admin and Analytics Support for managing application content and tracking searches, downloads, bookmarks and other application activity.
9. Advanced Features including product availability, advanced voice/image-assisted search, pricing updates, training certifications and approved calculators.
10. Source code, testing artifacts and final technical/project documentation.

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4. Success Criteria

The project will be evaluated against the following measurable criteria.

The primary quantitative success criterion specified in the project document is ≤3% MAPE on unseen tyre data, supported by robust group-wise validation.

No.	Success Criterion	Measurement
1	Mobile Application	Android and iOS application successfully functional
2	Product Catalogue	Products and technical information displayed correctly
3	Search & Filtering	Required search and filter flows work correctly
4	Localization	English, French, German and Italian validated
5	Customer Support	Support Library, FAQ, VOC and dealer support functional
6	Distributor Services	Role-based access and distributor features functional
7	Technical Tools	Required selectors and calculators functional
8	Analytics & Notifications	User analytics and push notifications operational
9	Production Deployment	Application successfully deployed to production
10	Testing	Final functional, UI and integration testing completed
11	Documentation	Technical and user documentation completed
12	Handover	Final project package and knowledge transfer completed

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5. Project Timeline

5.1 High-Level Timeline for the Project

Month	Activities
Month 1	Requirement analysis, API assessment, system architecture and development of core public-app features including product catalogue, search and support content.
Month 2	Completion of public features, backend/API integration, authentication and development of distributor-specific features such as asset library, training and claims.
Month 3	Development of advanced features and tools, analytics integration, system integration, testing, documentation and final project deliverables.

5.2 Detailed Date-Wise Project Timeline

Phase	Timeline	Major Activities
Requirement Analysis & Design	Week 1	Study requirements, perform API gap analysis and finalize system architecture.
Public App Development	Week 2–4	Product catalogue, search, support library, FAQs, testimonials, bookmarks, feedback and multilingual support.
Backend & Integration	Week 4–6	API/database integration, authentication, notifications, analytics and content integration.
Distributor Features	Week 6–8	Role-based access, asset library, training centre, claims portal and product-selection tools.
Advanced Features	Week 9–10	Availability, advanced search, pricing updates, certifications and approved calculators.
Integration & Documentation	Week 11–12	Module integration, testing, bug fixing and preparation of final documentation.

6. Key Milestones

Milestone	Target Date
Requirements & Existing App Assessment Complete	12-Aug-26
API Gap Analysis & Architecture Finalized	18-Aug-26
Launch 1 Feature Complete	24-Aug-26
Launch 1 Android Go-Live	25-Aug-26
Launch 2 Feature Complete	24-Sep-26
Launch 2 Production Release	25-Sep-26
Launch 3 Feature Complete	24-Oct-26
Launch 3 Production Release	25-Oct-26
Final Testing & Production Validation Complete	30-Oct-26
Final Project Handover	31-Oct-26

7. Key Risks and Mitigation

Risk	Mitigation
Incomplete product data	Data validation and content review
API gaps or instability	Early API assessment and fallback backend
Search/filter issues	Requirement-based search validation
Localization issues	Testing across all four languages
Access control issues	Role and permission validation
Calculator logic delays	Confirm formulas and inputs early
UI/device issues	Android and iOS device testing

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8. Final Completion Definition

The project will be considered complete when:

1. All three launch scopes have been implemented.
2. Product catalogue and search functionality are working.
3. Customer and distributor features are functional.
4. Required technical tools and calculators are integrated.
5. Backend, CMS, analytics, and Firebase services are operational.
6. Final testing and production deployment are completed.
7. Documentation and knowledge transfer are completed.
8. Final project handover has been conducted.

Final Target Completion Date: 31st October 2026

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9. Expected Business Outcome
   The completed solution is expected to improve customer and distributor engagement through a unified mobile platform, enabling faster product discovery, streamlined distributor workflows and more efficient access to product, pricing and support information. The project aims to reduce manual effort across customer and distributor interactions, improve sales responsiveness, strengthen data-driven decision-making through usage insights and create a scalable digital foundation for future AI-enabled customer and commercial capabilities.



Dr. Snehal Rathi



Dr. Viomesh Singh